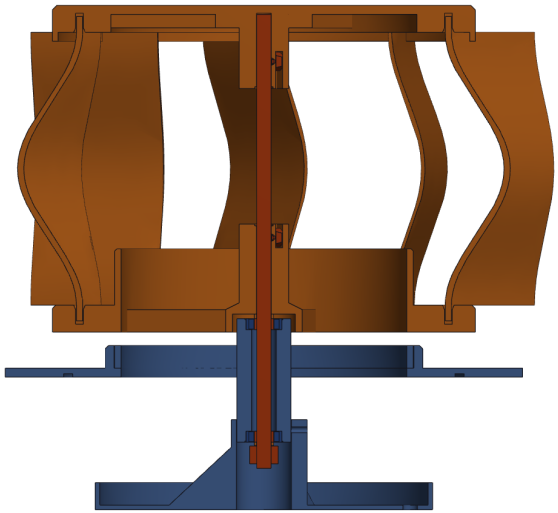
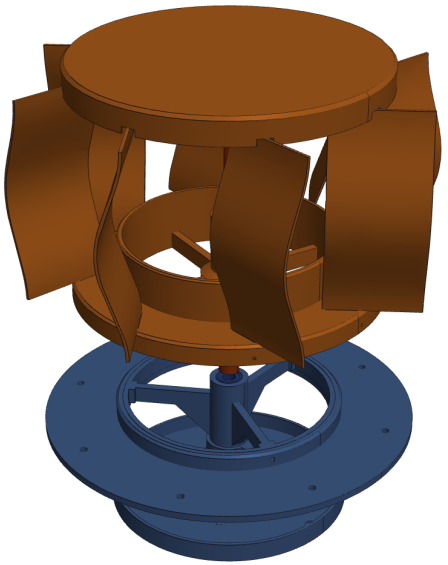
SHEET 3 - ROTOR vs STATIONARY: what turns, what is fixed to the roof, and how each attaches

SECTION - ORANGE rotating / BLUE stationary



Rotor (orange) lifted 70 mm off the stationary support (blue)



ROTATING part (orange)	Attaches to	By
Upper hub (closed top)	shaft (blind bore)	1x M4 set screw in nut slot -> shaft dimple
Lower hub (closed bottom)	shaft	1x M4 set screw in nut slot -> shaft dimple
8 blades	both hubs	tabs trapped in blind pockets (no fasteners)
Shaft 1/4 in	bearing INNER races	slip fit; lower hub bears on top inner race via washer
Top thrust washer	upper bearing inner race	loose on shaft, under lower-hub cup
Shaft collar + bottom washer	lower bearing inner race	collar set screw (uplift stop, 0.2 endplay)

STATIONARY part (blue)	Attaches to	By
Roof flash adapter	roof deck	8x #8 SS screws + sealant bead in groove
Bearing support (ring/spokes/tower)	adapter collar counterbore	rests on step; 3x M3 radial screws
2x R4-2RS bearing OUTER races	bearing support tower	pressed into seats (fit per coupon)
Rain tray (optional)	tower bottom (below roof)	slip-on standpipe + 1x M3 clamp screw

NO stationary part is fastened to either rotating hub. The only paths between rotor and stator are the two ball bearings. Minimum running clearances (from CAD): 6.0 mm lower hub to collar/support ring, 2.0 mm radial labyrinth (cup vs tower), 51 mm blade to blade, >13 mm blade to any stationary part.
Service: remove 3 M3 screws -> the whole rotor + bearing cartridge lifts out of the roof adapter.

DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Rotor vs stationary

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

SHEET 1/1 Source: build123d parametric model + MASTER_PARAMETERS.json